Chapter 1 INTRODUCTION TO STATISTICS

Q.1. Give the definition of statistics.

Ans. Statistics is defined as a mathematical science which is used for prediction and drawing conclusions in the situation of uncertainty. It includes designing of experiment, collection of data, presentation of data, analysis of data, interpretation of data and drawing inferences from the sample

Q.2. Define Statistics in three different ways?

Meanings of Statistics

- **Singular Sense**
- **Plural Sense**
- **Plural of word Statistic**

In Singular Sense, Statistics is the science of conducting studies to collect, organizes, summarize, analyze and draw conclusions from data.

In Plural Sense, The collection of numerical facts is called statistics e.g.; statistics of bowling in a cricket match.

Statistics as Plural of the term "Statistic" A numerical quantity computed from sample is called statistic. Two or more such measures are known as statistics

Q.2. How many are the branches of statistics? Explain?

Ans. There are two branches of statistics.

(i) Descriptive statistics

(ii) Inferential statistics

Descriptive statistics.

The branch of statistics that deals with collection, presentation and analysis of numerical data is called descriptive statistics (or deductive statistics).

Inferential statistics.

The branch of statistics that deals with drawing inferences about population on the basis of sample information is called inferential statistics (or inductive statistics).

Q.3. Define Population and sample.

Ans. Population: The total group under discussion or the group to which the results will be generalized is called population. For example, the set of all students in the college is our population.

Sample: A representative part of the population is called sample.

Q.4. Define a Ratio and proportion?

Ratio

The ratio of A to B is the fraction A/B. For example, if there are 30 females and 20 males in a party then male female ratio is 20/30.

Proportion

A proportion is a special ratio of a part to its total. For example, if there are 30 females and 20 males in a party then proportion of male and female is 20/50 and 30/50 respectively.

Q.5. Define Parameter and Statistic?

Parameter

A numerical quantity computed from population is called parameter. Parameters are constant and usually denoted by Greek letters. For example, population mean is denoted by μ and population standard deviation is denoted by σ.

Statistic

A numerical quantity computed from sample is called statistic. Statistics are variables because they vary from sample to sample. Statistics are denoted by Roman letters. For example, sample mean is denoted by X and sample standard deviation is denoted by S.

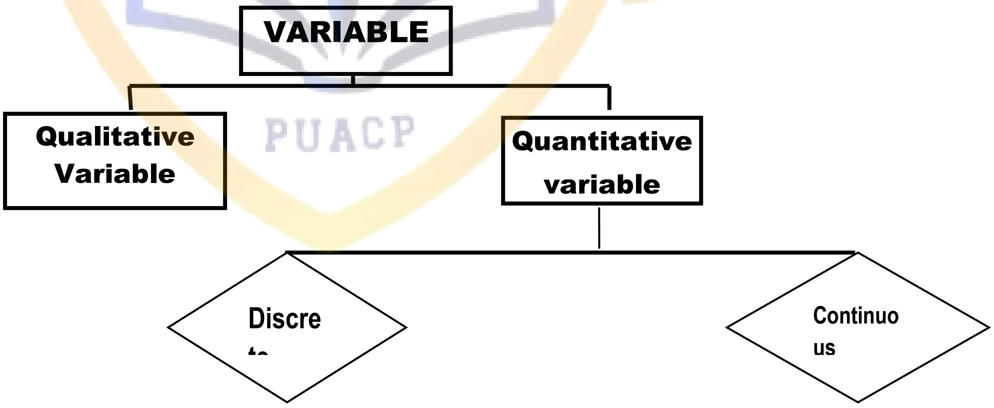
Q.6. What is difference between constant and variable?

Ans

Variable	Constant
A characteristic that varies from individual to individual or person to person is called variable. For example, weights of children, ages of students, temperature of a room, number of road accidents on motor way etc.	A characteristic that does not change from individual to individual or person to person but remains fixed is called constant. For example, the number of days in a week, the value of · = 22/7 etc.

Q.7. What are different types of variable? Explain each of them?

Ans.



There are two types are variables.

1. Quantitative Variable
2. Qualitative Variable

1. Quantitative Variable.

A variable which is capable of taking numerical value is called quantitative variable. For example, numbers of students in a class, height of a person, weight of a ball etc.

There are two types of Quantitative variable.

i. Discrete Variable **ii.** Continuous Variable

i. Discrete Variable.

A variable which can take only some specific values is called discrete variable. The examples of discrete variables are numbers of students in a class, numbers of persons in a city, number of buses in Lahore etc.

ii. Continuous Variable.

A variable which take any possible value within a given range is called continuous variable. The examples of continuous variable are weight of a ball, speed of a car etc.

2. Qualitative Variable.

Qualitative Variable is also called categorical variable. It is a variable, which is not capable of taking numerical variable. It is also called attribute. The examples of qualitative variables are intelligence, eye colour, sex, and beauty etc.

Q.8. Explain function (Uses) of statistics.

Ans.

(i) Statistics simplifies complexities.

(ii) Statistics presents facts in a definite form.

(iii) Statistics simplifies comparison of data.

(iv) Statistics studies relationship among different facts.

(v) Statistics studies change in the level of a given phenomenon.

(vi) Statistics aids forecasting.

Q.9. What does datum and data mean?

Ans. Datum

A single numerical fact is called datum

Data

The collection of information regarding any field of interest is called data. The plural of datum is data

Q.10. What is the difference between primary and secondary data?

Ans. There are two types of Data.

i. Primary Data: **ii.** Secondary Data

i. Primary Data.

Primary data is also called **Raw Data**. It is a data which has just been collected from the source and has not gone through any king of statistical treatment like sorting and tabulation.

ii. Secondary Data.

By secondary data we mean the data, which has already been collected by someone, that has undergone a statistical treatment like sorting and tabulation etc.

Q.12. What is measurement error?

Ans. The difference between the measured value and true value is called error of Measurement.

Q.13. Describe limitations of statistics.

Ans. Following are the some limitations of statistics.

(i) Statistics only deals with aggregates of facts.

(ii) Most of statistical techniques are useful for quantitative data only.

(iii) Statistical concepts are same for social and physical sciences.

(iv) Statistical results sometimes lead to confusion.

(v) Application of statistical techniques demands great expertise.

Chapter 2 Presentation of Data

Q.1. What is presentation of data?

Ans. The raw data, which have been collected, are usually very large in quantity. Therefore we have to organize and summarize the collected data in a form that is easy to understand. This is called presentation of data

Q.2. What are different methods of presentation of data?

Ans. Different methods of presentation of data are:

(i) Classification (ii) Tabulation
(iii) Diagrams (iv) Graphs

Q.3. Write a note on classification?

Ans. Classification is the process of arranging the data into relatively homogenous classes according to their resemblances and affinities.

There are different types of classification

One Way Classification

When data are classified by one variable, it is called one-way or simple classification. For example population of any location may be classified as rich, average income, poor according to the income level.

Two Way Classification

When data are classified by two variables at the same time, it is called two-way classification. For example we may classify the peoples on the basis of education and income group.

Manifold Classification

When data are classified by many variables, it is called manifold classification or cross classification. For example, we may classify the peoples on the basis of gender, education, income etc.

Q.4. Define the word tabulation?

Ans. The systematic arrangement of data in the form of rows and columns for the purpose of comparison and analysis is known as tabulation.

Following are types of tabulation:

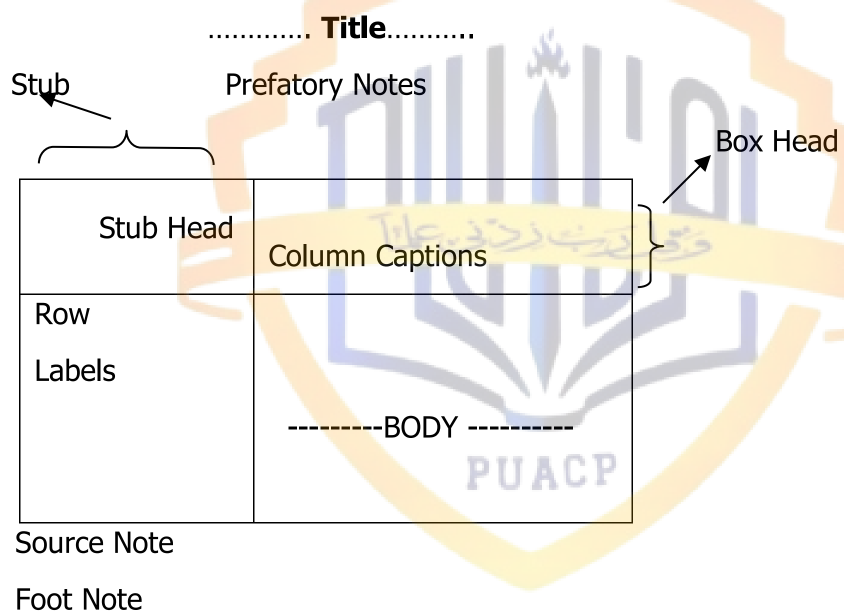
(i) Simple Tabulation (ii) Double Tabulation

(iii) Complex Tabulation

Q.5. What are the main parts of the table? Explain them?

Ans. Generally, a statistical table has the following parts:

(i) Title
(ii) Column Captions and Box-head
(iii) Row Captions and Stub
(iv) Body
(v) Prefatory Note
(vi) Foot Note
(vii) Source Note



Q.6. Define frequency distribution?

Ans. A frequency distribution is a tabular arrangement of the data in which various observations are arranged into classes along with their frequencies.

Q.7. What are class limits (or Inclusive Method)?

Ans. Class limit is also a technical term used to express non-overlapping classes. In this method both upper and lower class limits are included in the class interval. The classes in this design are technically known as "Class Limits".

Classes	10 – 19	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69

Q.8. What are class boundaries (or Exclusive Method)?

Ans. Class boundary is a technical term used to express overlapping classes. In this design lower limit is included but the upper limit is excluded from the class. The classes in this design are technically known as "Class Boundaries".

Classes	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 760

Q.9. What do mean by open-ended class?

Ans. If a frequency distribution has no lower limit or no upper limit of any class is called open-ended class.

Q.10. What do you mean by class marks (or mid points)?

Ans. A class mark is average value of the lower and upper class limits or class boundaries. It is also called as mid-pointor mid value; it may be denoted by X or Y.

Q.11. What is size of the class interval?

Ans. The difference between the upper and lower class boundaries is called size of the class interval or class width i.e. h = U.C.B - L.C.B.

Q.12. Define the class frequency?

Ans. The number of values falling in a specified class is called class frequency or frequency. It is denoted by f.

Q.13. Define the following?

Relative frequency, Percentage relative frequency, Cumulative frequency?

Ans. Relative Frequency

The frequency of a class divided by the total frequency of all the classes is called the relative frequency. i.e; Relative frequency = $\frac{f}{\sum f}$

The total of relative frequencies is unity (one). A table showing relative frequency is called relative frequency distribution.

Percentage Relative Frequency

If we multiply relative frequencies by 100, we obtain percentage relative frequency. A table showing percentage relative frequencies is called as percentage relative frequency distribution.

Cumulative Frequency

The total frequency of all the classes less than the upper class boundary of a given class is called as the cumulative frequency of that class.

Q.14. What is a graph? Write down names of graphs of frequency distribution?

Ans. Graph represents the data in the form of lines or curves to show the fluctuations, trends and other features of the data.

The graphs of frequency distribution are given below:

(i) Histogram (Frequency Histogram) (ii) Polygon (Frequency Polygon)

(iii) Curve (Frequency Curve) (iv) Ogive (Cumulative Frequency Polygon)

Q.15. What is histogram?

Ans. A frequency histogram or histogram is a set of adjacent rectangles for a frequency distribution such that the area of each rectangle is proportional to the corresponding class frequency.

We take the class boundaries (C.B's) along X-axis and frequency (f) along Y-axis. If the class intervals are not equal, the height of the rectangle over an unequal class interval is to be adjusted.

Q.16. What is historigram?

Ans. The graph of a time series is called historigram. It is constructed by taking time along x-axis and the values along y-axis. Using an appropriate scale, points are plotted and then these points are joined by line segments to get required historigram.

Q.17. What is frequency polygon?

Ans. A frequency polygon is a closed geometric figure used to display a frequency distribution graphically. Following steps are taken to make a frequency polygon from a frequency distribution.

(i) Calculate mid values of the class boundaries.
(ii) Mark these mid values along x-axis.
(iii) Mark the frequencies along y-axis.
(iv) Mark corresponding frequencies against each mid-point, join them and extend it to x-axis.

Q.18. What is a frequency curve and ogive?

Ans. **Frequency Curve**

When a frequency polygon is smoothed, it approaches a continuous curve called a frequency curve.

Ogive

An ogive or cumulative frequency polygon is a graph obtained by plotting the cumulative frequencies of a distribution against the upper or lower class boundaries depending upon "less than" or "more than" type and points are joined by line segments.

Q.19 What is a chart? Write names of different charts?

Ans. A chart (or diagram) is a device used for representing a simple statistical data in a simple, clear and effective manner.

Name of some diagrams (or charts).

Simple bar chart, sub-divided bar chart, multiple bar chart, component bar chart, percentage component bar chart, pie chart (diagram).

Q.20. Define simple bar diagram?

Ans. A bar diagram representing a single item of data is called a simple bar diagram (chart) e.g. Bar chart showing total population of Pakistan in selected years. Bar chart showing number of workers in different industries of Pakistan in 1995, and so on.

Q.21. What are multiple bar charts?

Ans. In multiple bar charts, two or more characteristics related to the values of a common variable in the form of grouped bars are drawn. The height of the bar represents the actual value of the characteristics. For example, male and female population of Pakistan in the census reports can be shown by separate bars for male and female population placed adjacent to each other for each of the years.

Q.22. What is the difference between component and percentage component bar chart?

Ans. **Component Bar Chart**

A component bar chart (or Sub-divided bar chart) is a technique in which each bar is divided into two or more sections proportional in size to the component parts of a total being displayed by each bar.

Percentage Component Bar Chart

The interest may sometimes be not on the increase or decrease in absolute values, but on the change in the relative share of each part to the total. To present this, the bars are divided in percentages. All bars will be of the same length or height equal to 100. This presentation is known as percentage component bar chart.

Q.23. Differentiate between pictogram and pie diagram?

Ans. **Pictogram**

A pictogram is a popular device for displaying the statistical data by means of pictures or small symbols.

Pie Chart

A pie chart is a diagrammatic representation of data in circular form. In a pie chart, a circle is divided into a number of sectors. The area of a sector of a circle is proportional to the angle of the sector; the angle can be calculated as

$$\theta = \frac{\text{Component Part}}{Total} \times 360$$